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St. JOSEPH'S COLLEGE OF ENGINEERING

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OMR, CHENNAI - 119

DEPARTMENT OF INFORMATION TECHNOLOGY

AD1006 - Unnat Bharat Abhiyan (UBA)

उन्नत भारत अभियान

UNNAT BHARAT ABHIYAN

Focusing Area: Medical & Health.

Location: Kolivakkam village, Kancheepuram District

REPORT

Submitted by

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UBA FOCUSING AREA MAPPING WITH SDG AND JUSTIFICATION

SDG Goal 3: Ensure healthy lives and promote well-being for all at all ages

This project, *"Developing a community based health monitoring system for early detection non-communicable disease,"* directly addresses the healthcare access challenges faced by rural populations through the integration of telemedicine services. Implemented in Kolivakkam village, the initiative aims to provide equitable and timely medical support, especially to underserved groups such as women, children, the elderly, and people with chronic illnesses.

By leveraging digital technology and community health worker support, the project ensures accessible, affordable, and continuous healthcare services in remote locations, contributing to better health outcomes and reducing the rural-urban healthcare divide.

Relevant Targets and Contributions:

- **3.8 – Achieve universal health coverage, including financial risk protection, access to quality essential healthcare services, and access to safe, effective, quality, and affordable essential medicines and vaccines for all.**
  The project ensures access to quality healthcare through virtual consultations, reducing the need for travel and high expenses while providing timely medical advice and follow-ups.
- **3.c – Substantially increase health financing and recruitment, development, training, and retention of the health workforce in developing countries.**
  Through the training of local healthcare workers in telemedicine tools and virtual patient care, the project builds the capacity of the rural health system and strengthens the local workforce.

UBA SEG - Project proposal format

1. **Title of the Technology**

   Developing a Community-Based Health Monitoring System for Early Detection of Non-Communicable Diseases (NCDs) in Kolivakkam village, Kancheepuram District

2. **Background of the Project**

Introduction

Kolivakkam village, located near the historic town of Kanchipuram in Tamil Nadu, faces significant challenges in the domain of preventive healthcare. One of the most pressing concerns is the increasing prevalence of non-communicable diseases (NCDs) such as diabetes, hypertension, cardiovascular conditions, and obesity-related issues. These conditions often go undiagnosed or are detected only at advanced stages due to a lack of regular screening, low health literacy, and inadequate health infrastructure. The village does not have a consistent mechanism for community-level health monitoring, and the outreach of Primary Health Centres (PHCs) is often limited by logistical constraints, manpower shortages, and outdated methods of tracking patients.Furthermore, frontline health workers such as ASHA and Anganwadi workers operate with minimal digital support and often lack access to real-time patient data. This gap not only limits their ability to provide timely interventions but also weakens the overall healthcare response at the grassroots level. Cultural stigma and poor awareness about the long-term effects of NCDs also discourage people from seeking preventive care. There is an urgent need to bridge the knowledge gap, enhance screening frequency, and empower health workers with modern tools to better serve the community.

Socio-Economic Context and Cultural Context

Kolivakkam village reflects a typical rural socio-economic structure, where agriculture and manual labor are the main sources of income. Limited financial resources and inadequate infrastructure hinder access to quality healthcare and education. Culturally, the village is rooted in traditional beliefs, with many still relying on home remedies or local healers. Gender roles significantly influence healthcare decisions, often restricting women's and elderly people's access to medical services. Awareness about preventive healthcare and non-communicable diseases remains low within the community.

Infrastructure and Access

Access to Healthcare Tools and Equipment partnerships with local hospitals, medical colleges, and healthcare NGOs will help ensure access to the necessary medical tools for screenings and diagnostics. Additionally, collaboration with digital health companies will provide the technology required for mobile health tracking and data collection.

Access to Trained Health Workers access to trained healthcare professionals, including ASHA workers, nurses, doctors, and volunteers, is essential. These workers will require continuous training and skill development in using mobile health tools, conducting screenings, and educating the community on preventive healthcare.

Environmental and Policy Relevance

The Community Health Watch project holds significant environmental and policy relevance by aligning with national goals for sustainable health practices and addressing critical public health challenges. Environmentally, the project promotes preventive healthcare, encouraging healthier lifestyles that can reduce the reliance on medical treatments and minimize the environmental impact associated with healthcare waste and resource consumption.

Proposed Solution

The proposed solution for the Community Health Watch project involves the establishment of a grassroots-level health monitoring system designed to empower rural communities, particularly in Kolivakkam village, with the tools and knowledge necessary to prevent and manage non-communicable diseases (NCDs). This will be achieved through the deployment of mobile health units that will travel across the village to conduct regular health screenings for conditions such as diabetes, hypertension, and cardiovascular diseases. These units will be equipped with basic diagnostic tools like blood pressure cuffs, glucometers, and pulse oximeters

Expected Impact

The Community Health Watch project is expected to have a transformative impact on both individual health outcomes and the broader community in Kolivakkam village.Firstly, by providing regular health screenings and data-driven monitoring, the project aims to significantly reduce the prevalence of undiagnosed non-communicable diseases (NCDs) such as diabetes, hypertension, and cardiovascular conditions. Early detection will enable timely intervention, potentially preventing the progression of these diseases and improving long-term health outcomes for individuals.

Implementation Strategy

The first phase will focus on engaging the community and raising awareness about the project. This will involve meetings with local leaders, ASHA workers, and community groups to introduce the project's objectives and the benefits of preventive healthcare. Awareness campaigns, including posters, local events, and workshops, will be organized to educate the villagers about non-communicable diseases (NCDs) and the importance of early detection.

In this phase, the necessary infrastructure will be established. This includes setting up mobile health units equipped with diagnostic tools (such as blood pressure monitors, glucometers, and pulse oximeters). The mobile health units will be stationed in accessible locations across the village for easy reach by the community. Additionally, a mobile health tracking platform will be developed to enable ASHA workers and volunteers to collect and monitor health data.

Conclusion

To foster a healthier and more resilient future, interventions in rural healthcare systems must prioritize accessibility and early intervention. The Community Health Watch project aims to bridge the gap in preventive healthcare in Kolivakkam village by empowering the community with tools and knowledge to manage and prevent non-communicable diseases. The outcome will not only be improved health outcomes but also a model for sustainable, community-driven healthcare that can be replicated across rural India, paving the way for healthier, more informed communities.

3. **Brief Objective/s of the Project** :

The primary objective of the Community Health Watch – A Rural NCD Screening and Awareness Model project is to enhance healthcare access and improve the early detection and management of non-communicable diseases (NCDs) in Kolivakkam village. The initiative focuses on empowering rural communities with the knowledge, tools, and support to take charge of their health and prevent the onset of chronic diseases such as diabetes, hypertension, and cardiovascular conditions. This project is designed to address critical healthcare challenges, promote preventive health practices, and create a replicable, sustainable model for rural health empowerment. The following objectives outline the core goals of the project:

1. **Improving Health Awareness and Education**

A major objective of the project is to increase health awareness and education within the community. This will be achieved through regular health awareness workshops and outreach campaigns aimed at educating villagers about lifestyle changes, healthy eating, physical activity, and the importance of regular health check-ups. Alongside workshops, informational materials such as posters and pamphlets will be distributed to raise awareness about the risks associated with common NCDs and the preventive measures that can be taken to avoid them. By disseminating knowledge about the key risk factors for NCDs and the steps that can be taken to mitigate them, the project aims to equip the population with the tools to improve their long-term health.

2. **Strengthening Healthcare Access**

A crucial aspect of the project is strengthening healthcare access, particularly routine health screenings and NCD management services. To achieve this, the project will organize periodic health camps at various locations within the village to provide free screening for NCDs, including diabetes, hypertension, and cardiovascular diseases. These camps will help identify undiagnosed conditions and enable early intervention. The project will also provide basic diagnostic tools such as blood pressure monitors, glucometers, and pulse oximeters to local healthcare workers, ensuring they have the necessary tools to effectively screen and track health data. Additionally, mobile health units will be established to reach underserved areas within the village, ensuring no one is excluded from healthcare access due to geographical barriers.

3. Empowering Local Healthcare Workers

Another important goal of the project is to empower local ASHA workers and healthcare volunteers, enhancing their capacity to monitor and support NCD patients. Training sessions will be conducted for ASHA workers and healthcare volunteers, teaching them how to properly use diagnostic tools, enter data, and follow up with patients. Additionally, these workers will be trained in interpreting health data, offering lifestyle modification advice, and providing guidance on managing chronic diseases. A mobile health platform will be developed for real-time health data collection and tracking, which will help healthcare workers provide personalized care and support to individuals diagnosed with NCDs.

4. Leveraging Technology for Healthcare Monitoring

Technology will play a key role in the project, particularly in health data monitoring. A mobile health tracking platform will be used to collect and monitor health data, enabling healthcare workers to track patients' progress, send reminders for follow-up appointments, and intervene when necessary. Community members will also have access to their health records and follow-up instructions via mobile applications or SMS, ensuring that patients remain engaged in their healthcare and adhere to prescribed management plans. This digital platform will make healthcare more accessible and transparent for the residents of Kolivakkam, facilitating a more efficient and streamlined health monitoring process.

5. Community Engagement and Participation

To ensure the success of the project, community engagement and participation will be encouraged. The project will involve local leaders, teachers, and other community members in promoting health screenings, wellness activities, and educational programs. Community meetings will be organized to motivate residents to participate in health screenings, while awareness campaigns will highlight the importance of NCD prevention. The active involvement of the community is critical in fostering a culture of health and well-being and in ensuring that the project is embraced and sustained by the villagers.

6. Promoting Sustainable Health Practices

A key aspect of the project is its focus on promoting sustainable health practices. To ensure that the initiative is environmentally friendly and energy-efficient, solar-powered mobile health units will be utilized. Additionally, low-cost, durable diagnostic kits will be designed to ensure the project is affordable and scalable. The project will also create open-source health education materials and monitoring guidelines that can be used by other rural communities. By using these low-cost solutions and promoting the use of solar energy, the project will be able to offer a sustainable healthcare model that can be replicated in other underserved areas.

Conclusion

In conclusion, the Community Health Watch – A Rural NCD Screening and Awareness Model project is designed to address the critical need for accessible healthcare in rural areas, particularly in Kolivakkam village, where preventive care and early detection of NCDs are often limited. By providing mobile health units, training local healthcare workers, and engaging the community in health awareness programs, this project aims to create a sustainable, community-driven model that empowers rural populations to manage their health effectively. Through regular health screenings, data-driven monitoring, and educational initiatives, the project will reduce the incidence of undiagnosed NCDs, improve overall health outcomes, and create a foundation for better healthcare in rural India. Furthermore, the approach will serve as a scalable model, enabling other underserved communities to replicate the project and benefit from its impact, ultimately contributing to a healthier, more informed nation.

7. Community Engagement and Needs Assessment

- Effective community engagement and a thorough needs assessment are foundational to the success of the *Community Health Watch* project in Kolivakkam village. The initiative begins by building trust and rapport with the residents through regular interactions, village meetings, and collaboration with local leaders, teachers, and healthcare workers. Understanding the unique socio-cultural dynamics of Kolivakkam is essential for designing an intervention that resonates with the community's values and everyday realities.
- A comprehensive needs assessment will be conducted to identify gaps in existing healthcare infrastructure, community awareness about non-communicable diseases

(NCDs), and the barriers faced by residents in accessing preventive health services. This will include household surveys, focus group discussions with villagers, and consultations with ASHA workers and local clinics. These efforts will provide critical insights into the prevalence of conditions like diabetes and hypertension, current health-seeking behaviors, and the level of trust in the local healthcare system.

- Community engagement doesn't end with initial assessments—it will be an ongoing process throughout the project. Local youth and women will be encouraged to take active roles as health ambassadors, spreading awareness and helping organize screening camps and health awareness drives. By involving the community directly in the planning, implementation, and monitoring stages, the project fosters a sense of ownership, making the intervention more sustainable and impactful in the long term. This participatory approach ensures that the solutions provided are not only need-based but also culturally acceptable and community-driven.

8. Curriculum Development and Training Design

- The *Community Health Watch* project in Kolivakkam village emphasizes the development of a well-structured curriculum and training design tailored specifically to the rural healthcare context. The curriculum is centered around non-communicable disease (NCD) awareness, early identification, and effective community-based health management strategies. It is designed to be simple, visual, and accessible, ensuring that even individuals with limited formal education, such as ASHA workers and local volunteers, can understand and apply the knowledge effectively.
- Training modules will be developed in the local language (Tamil) and will include topics such as the basics of NCDs, symptoms to watch for, preventive measures, lifestyle changes, and the importance of early screening. The curriculum will also incorporate practical sessions on using mobile-based health monitoring tools for data collection and record keeping. This training will equip frontline health workers with the technical and interpersonal skills required to interact confidently with villagers, conduct screenings, and promote healthy habits.
- In addition, periodic refresher workshops and digital learning materials will ensure continuous learning and skill enhancement. Role-plays, case studies, and visual demonstrations will be used to make training interactive and relatable. Special focus will be given to empowering women volunteers, enabling them to become community health champions. Through this

targeted and inclusive training approach, the project aims to build a competent local health workforce that can sustain preventive healthcare efforts in Kolivakkam long after the initial implementation phase.

9. Training and Capacity Building

1.Training and capacity building form the core of the *Community Health Watch* project, ensuring that local health workers and volunteers in Kolivakkam village are well-equipped to carry out NCD screening, health education, and community engagement activities. The project will provide intensive, hands-on training to ASHA workers, Anganwadi staff, and selected community volunteers, empowering them with both technical knowledge and practical skills in preventive healthcare.

2.The training program will be conducted in multiple phases, beginning with a foundational module covering the basics of non-communicable diseases such as diabetes, hypertension, and cardiovascular conditions. Participants will learn how to recognize early warning signs, use diagnostic tools like glucometers and digital blood pressure monitors, and record and report health data using mobile applications designed for rural settings. Training will also include communication skills, counseling techniques, and methods for organizing community health events.

3.Workshops will be interactive and culturally contextualized, utilizing storytelling, role-playing, and visual demonstrations to reinforce learning. To maintain long-term engagement, the project will incorporate regular refresher sessions and peer-to-peer learning opportunities. Expert trainers, including healthcare professionals and public health educators, will mentor the trainees, offering guidance and addressing challenges they face in the field.

10. Mentorship and Exposure Opportunities

- The *Community Health Watch* project recognizes the transformative potential of mentorship and exposure in empowering rural health workers and volunteers. In Kolivakkam village, creating connections between local community members and experienced professionals in the fields of healthcare, public health, and community development will be a key strategy to inspire learning, build confidence, and promote sustainability.
- A structured mentorship program will be established where experienced medical practitioners, nurses, public health experts, and social workers will be invited to guide the trained ASHA

workers and volunteers. These mentors will share best practices, provide professional insights, and offer ongoing advice through both in-person visits and virtual sessions. Regular interactions will help the local health workers stay updated with new knowledge and feel supported in their roles.

- In addition, exposure visits will be arranged for selected community health workers to model health facilities, public health campaigns, and successful community-driven healthcare projects within the district or nearby regions. These visits aim to broaden their perspectives, enhance their problem-solving abilities, and help them understand how scalable models can be implemented locally.
- Workshops and networking sessions with healthcare innovators and NGOs will also be organized to expose local workers to digital health tools, community outreach techniques, and sustainable practices. By integrating mentorship and exposure opportunities into the training process, the project will foster a motivated and knowledgeable grassroots workforce capable of independently leading Kolivakkam village towards better health outcomes.

11. Monitoring and Evaluation

- The *Community Health Watch* project places strong emphasis on a robust Monitoring and Evaluation (M&E) framework to ensure transparency, effectiveness, and continuous improvement throughout its implementation in Kolivakkam village. This framework will guide the assessment of activities, track progress toward goals, and inform decision-making to maximize the project's impact.
- From the outset, baseline surveys will be conducted to gather detailed data on the health status, awareness levels, and access to screening services in the community. These surveys will help identify gaps, set measurable targets, and establish a reference point for evaluating changes brought about by the project.
- Ongoing data collection will be performed using mobile-based health monitoring tools provided to ASHA workers and volunteers. These tools will log screening data, health indicators, and outreach activity records in real-time, enabling timely analysis. Periodic reviews will be held to assess the performance of field workers, participation levels of community members, and the functionality of the digital tracking systems.
- Qualitative assessments such as focus group discussions, interviews, and community feedback sessions will offer insights into behavioral changes, satisfaction levels, and cultural

acceptability of the intervention. These findings will help tailor the program to better suit local needs and preferences.

- At the conclusion of each project phase, a comprehensive impact evaluation report will be compiled. This report will highlight successes, lessons learned, and recommendations for replication in other rural areas. By embedding monitoring and evaluation into every stage, the project ensures accountability and adaptability, leading to a scalable and sustainable health intervention model for Kolivakkam and beyond.

12. Sustainability and Follow-Up Support

4. Ensuring the long-term sustainability of the *Community Health Watch* project in Kolivakkam village is a key priority. The initiative has been carefully designed to build local capacity, foster ownership within the community, and create mechanisms that will endure beyond the project's active implementation phase.
5. To achieve sustainability, the project focuses on empowering local stakeholders—especially ASHA workers, community volunteers, and school teachers—with the skills, tools, and confidence required to manage health awareness and screening independently. Regular training, refresher courses, and access to open-source digital health tools will enable these individuals to continue the work without ongoing external support.
6. In addition to individual empowerment, the project will work closely with local Panchayat leaders and health officials to integrate the initiative into existing government health programs and policies. By aligning with the goals of the National Health Mission and state-level rural health schemes, the project seeks to secure long-term institutional support and potential funding for its continuation.
7. Follow-up support will be provided in the form of mentorship from medical professionals and NGOs, who will remain connected to the community through quarterly virtual check-ins and on-demand consultations. A local monitoring committee composed of village representatives and health workers will be formed to oversee activities, track progress, and address emerging challenges.

Conclusion

The *Community Health Watch – A Rural NCD Screening and Awareness Model* in Kolivakkam village represents a thoughtful and transformative approach to bridging healthcare gaps in underserved rural areas. By focusing on the early detection and management of non-communicable diseases (NCDs), the project addresses a growing concern that often goes unnoticed due to lack of access, awareness, and infrastructure.

Through the strategic integration of mobile technology, grassroots participation, and female-led healthcare delivery, the initiative empowers ASHA workers and community volunteers to become catalysts for health-driven change. The project's structure—spanning curriculum development, capacity building, mentorship, and strong monitoring—ensures that the community is not only engaged but also equipped to sustain the benefits independently over time.

10. Name of Principal Investigator (25 word) Dr.M.Usha
11. E-mail of Principal Investigator * Usha.m@kgcas.com
12. Mobile No. of Principal Investigator 94877 90087

Total cost of the Product / Technology	
Categories of Fund	**Amount**
Mobile Components (Science + Coding)	Rs.35,000
Training and Honorarium for Female Mentors	Rs.20,000
Transportation and Outreach (Rural Access)	Rs.10,000
Digital Devices / Tablets (shared use)	Rs.15,000
Running Cost Printed Learning Materials & Stationery	Rs.5,000
Workshop Setup (Tents, Furniture, etc.)	Rs.10,000
Miscellaneous / Contingency	Rs.5,000
Total Proposed Amount	**Rs.100,000**

13. Funds Raised

The *Community Health Watch* project has garnered support through a combination of institutional grants, corporate social responsibility (CSR) contributions, and community-based fundraising efforts. Initial funding was secured through the **Unnat Bharat Abhiyan (UBA)** program, which provided foundational support for planning, research, and pilot activities.

Additionally, outreach to health-focused CSR partners and philanthropic organizations helped raise further capital to procure medical equipment, mobile screening kits, and digital health tools essential for field operations. Partnerships with local NGOs and healthcare companies enabled in-kind donations, such as glucometers, blood pressure monitors, and awareness materials, significantly reducing equipment costs.

Community involvement played a vital role in fundraising, with local panchayats and stakeholders contributing space, volunteers, and logistical support. Educational institutions also aided in mobilizing resources and volunteers through student-led health drives and awareness campaigns.

Overall, the combination of structured institutional backing and grassroots participation has ensured the financial viability of the project in its early phases, while also laying the groundwork for long-term funding through government health schemes and sustainability grants.

1. Initial Planning and Community Engagement

The project begins with thorough planning, including stakeholder identification and community engagement. Initial meetings will be held to introduce the project, gather insights, and foster a sense of ownership among the tribal women and local leaders. A needs assessment survey will be conducted to collect quantitative and qualitative data on the socio-economic conditions of the community. This assessment will guide the

development of a tailored training curriculum that addresses the specific needs and aspirations of the participants.

2. Curriculum Development

Based on the findings from the needs assessment, a comprehensive training curriculum will be developed. This curriculum will focus on practical skills for biodegradable product development using local materials such as banana fibers, paddy straw, and sugarcane extracts. The curriculum will include modules on sustainable practices, product design, and business management. Input from local experts and trainers will be integrated to ensure the content is relevant and effective.

3. Training Implementation

The training phase will involve a series of workshops designed to provide hands-on experience. Participants will learn techniques for extracting and processing fibers, creating biodegradable products, and managing small businesses. Each workshop will be interactive, encouraging participants to practice the skills they acquire. To enhance learning, mentorship programs will be established, pairing participants with experienced mentors who can provide guidance and support throughout the training process.

4. Market Integration and Business Development

As participants develop their skills, the project will facilitate their integration into local and regional markets. Market research will be conducted to identify potential buyers and pricing strategies for biodegradable products. Business planning workshops will guide participants in creating individual business plans, covering essential topics such as marketing, branding, and financial management. The project will also organize opportunities for participants to showcase their products at local fairs and exhibitions, helping them establish a customer base.

5. Monitoring and Evaluation

To ensure the project is on track and making the desired impact, a robust monitoring and evaluation framework will be established. Baseline data will be collected prior to implementation, allowing for the assessment of changes in participants' socio-economic status and skills. Regular monitoring will involve feedback sessions with participants and stakeholders, enabling timely adjustments to training and support services.

6. Sustainability Planning

Sustainability is a key focus throughout the project execution. To promote long-term success, participants will be encouraged to form cooperatives or self-help groups that foster collaboration in production and marketing. Ongoing training sessions will be organized to address advanced topics and adapt to market changes. Networking opportunities with other entrepreneurs and support organizations will further strengthen the participants' capacity to sustain their businesses.

7. Final Reporting and Documentation

Upon completion of the project, comprehensive reports will be prepared to document outcomes, success stories, and lessons learned. This documentation will serve as a valuable resource for future initiatives and provide insights into effective practices for empowering tribal women and promoting sustainable development.

14. Impact on village/ Beneficiaries

The *Community Health Watch* project is expected to bring a profound and measurable impact on the lives of residents in Kolivakkam village. By focusing on early screening and awareness of non-communicable diseases (NCDs), the project aims to shift the healthcare approach from reactive to preventive, which is vital in reducing the long-term burden of chronic illnesses.

For the villagers, especially middle-aged and elderly individuals who are most vulnerable to diabetes, hypertension, and cardiovascular issues, this initiative will provide regular health check-ups that were previously inaccessible or unaffordable.

1. Initial Planning and Community Engagement

The foundation of the *Community Health Watch – A Rural NCD Screening and Awareness Model* was laid through detailed initial planning and deep-rooted community engagement in Kolivakkam village. The planning phase involved identifying key health challenges faced by the village, with a specific focus on the rising prevalence of non-communicable diseases (NCDs) like diabetes and hypertension. To understand the community's existing healthcare practices, a situational analysis was conducted through surveys, discussions with ASHA workers, and consultations with local health officials.

Community engagement played a pivotal role in ensuring the relevance and acceptance of the project. Village meetings were organized to introduce the initiative, gather feedback, and identify key concerns. These dialogues helped build trust and ensured that the community felt a sense of ownership over the program from the start.

2. Curriculum and Kit Development

The development of a tailored curriculum and health screening kits was a critical aspect of the *Community Health Watch* project. The curriculum was designed to address the specific needs of the Kolivakkam community, focusing on the early identification and management of non-communicable diseases (NCDs) such as diabetes, hypertension, and cardiovascular conditions. The curriculum was built to be simple, engaging, and culturally relevant, ensuring it could be easily understood and adopted by community members.

3. Mentorship and Life Skills Integration

A core component of the *Community Health Watch* project is the integration of mentorship and life skills training, which plays a pivotal role in empowering the community, especially women and youth, to take charge of their health and well-being. Mentorship not only provides guidance on health-related issues but also aims to develop crucial life skills that enable individuals to make informed decisions, foster personal growth, and become active contributors to their community's overall health.Mentorship sessions were structured to involve local female healthcare professionals and experienced community leaders who could serve as role models.

4. Final Documentation and Knowledge Sharing

Upon completion, the outcomes, impact metrics, and stories of change will be compiled into a detailed report. The documentation will include case studies, video testimonials, and an implementation toolkit that can be replicated in other villages. The findings will be shared at educational forums and with policy-makers to promote scaling of the initiative.

15. Duration of Implementation of Project: 6 months

In the first 6 months of the *Community Health Watch* project, the focus will be on foundational activities that set the stage for long-term success. The initial two months will be dedicated to thorough planning and engaging with the community to understand their health needs, build trust, and ensure active participation. During months 3 and 4, the development of health education curricula and mobile screening kits will be completed, with training sessions for ASHA workers and volunteers. In months 5 and 6, the mentorship program will be introduced, alongside life skills training, empowering community members to take a proactive role in their health. By the end of the 6-month mark, the project will have established a solid foundation of education, training, and community engagement, ready for the implementation of health screenings and awareness campaigns in the following months.

16. How to maintain future sustainability of installed technology in the village 500 words left

Ensuring the future sustainability of the mobile and digital learning infrastructure in Agaram village is essential for long-term impact. A robust sustainability plan that includes community participation, teacher involvement, continuous training, equipment maintenance, and institutional support will ensure that the technology remains functional and impactful even after the project's initial phase. The following key strategies will help in maintaining and enhancing the sustainability of the initiative:

1. Community Ownership and Engagement

Building a sense of ownership among local stakeholders is crucial to long-term success.

- **Inclusive Participation**: Involve school teachers, parents, and local leaders in decision-making related to the use and scheduling of the mobile lab and resources. This promotes accountability and shared responsibility.

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2. Capacity Building and Training

Sustainability depends on regular training and development of both students and teachers.

- **Teacher Training Programs**: Conduct refresher training for science teachers on how to facilitate hands-on experiments, coding activities, and mentorship integration using the kits.
- **Peer Mentorship**: Establish a peer learning model where trained students can guide new learners, ensuring knowledge transfer across batches.
- **Digital Literacy Camps**: Periodic sessions will be conducted to keep students

updated on new tools, platforms, and digital skills

3. Maintenance and Support Systems

- **Regular Equipment Checks**: Schedule monthly check-ups for science and coding kits, tablets, and mobile lab infrastructure to ensure everything is in working condition.

- **User Manuals and Maintenance Logs**: Provide user-friendly guides and assign student or teacher leads to log equipment usage and maintenance issues.

4. Institutional Integration and Educational Alignment

Embedding the mobile lab into existing school programs ensures it becomes a core part of the academic routine.

- **Funding through CSR and NGOs**: Seek long-term collaborations with educational foundations, tech companies, and government bodies for kit replacement, mentor support, and digital upgrades.

- **Alumni Engagement**: Engage former students as volunteers or peer mentors, building a sustainable cycle of mentorship and role modeling.

5. Monitoring and Feedback Systems

Tracking progress and adapting the initiative as needed ensures ongoing relevance and improvement.

- **Impact Indicators**: Collaborate with local government officials to advocate for policies that support women's entrepreneurship and sustainable practices. Define measurable indicators such as participation rate, concept retention, skill improvement, and interest in careers.

- **Feedback Channels**: Collect structured feedback from students, teachers, and parents to make iterative improvements in sessions, tools, and teaching approaches.

Conclusion

The *Community Health Watch – A Rural NCD Screening and Awareness Model* in Kolivakkam village represents a thoughtful and transformative approach to bridging healthcare gaps in underserved rural areas. By focusing on the early detection and management of non-communicable diseases (NCDs), the project addresses a growing concern that often goes unnoticed due to lack of access, awareness, and infrastructure.

Through the strategic integration of mobile technology, grassroots participation, and female-led healthcare delivery, the initiative empowers ASHA workers and community volunteers to become catalysts for health-driven change. The project's structure—spanning curriculum development, capacity building, mentorship, and strong monitoring—ensures that the community is not only engaged but also equipped to sustain the benefits independently over time.

17. Impact of this work on learning of Teacher

The *Community Health Watch* project will have a significant impact on the learning and professional development of the teachers involved in the initiative. Through their engagement with the project, teachers will gain practical knowledge and experience in health education, preventive care, and community-based health promotion. They will be exposed to a new curriculum designed specifically for rural health awareness, which will not only enhance their teaching abilities but also equip them with valuable skills in integrating health education into their regular classroom activities.

Teachers will also benefit from exposure to mentorship programs and workshops conducted by health professionals, which will broaden their understanding of the socio-economic and health challenges faced by rural communities.

18. Role of PI after compilation of the project duration

After the completion of the *Community Health Watch* project's implementation phase, the Principal Investigator (PI) will continue to play a critical role in ensuring the sustainability and continued success of the project's outcomes. The PI will be responsible for overseeing the final stages of project evaluation, which includes reviewing the impact assessment reports and ensuring that all findings are accurately documented. This will involve analyzing data collected throughout the project to measure its effectiveness in improving health outcomes, empowering the community, and fostering long-term health behavior changes.

Additionally, the PI will work closely with local stakeholders, including healthcare professionals, community leaders, and ASHA workers, to transition the project to a self-sustaining model. This may involve providing further training or technical support to ensure that the community can independently manage the health monitoring system and awareness programs.

19. Duration of monitoring by PI Post Completion of the project

The Principal Investigator (PI) will monitor the project for **6 months** after its formal completion. During this period, the PI will oversee the continued use of the mobile mentor sessions, and student participation . Regular visits to the village will be scheduled to observe teaching practices, gather feedback from students and teachers, and assess the project's integration into school routines. The PI will also track the functionality of the equipment and any technical or training needs. This ongoing support will ensure that the project's benefits are sustained and that improvements can be implemented when necessary.